# BAHIRDAR UNIVERSITY

# BAHIRDAR INSTITUTE OF TECHNOLOGY (BIT)

# FACULTY OF COMPUTING

# SOFTWARE ENGINEERING

## FINAL YEAR PROJECT TITLE SUBMISSION


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# Project Title 1 Secure Escrow and E-Commerce Mediation System

**Introduction**

E-commerce in Ethiopia is rapidly expanding through social media and informal digital marketplaces. However, most transactions require buyers to pay sellers directly without delivery guarantees. The absence of a secure payment mediation mechanism exposes both parties to fraud, disputes, and financial loss, limiting trust and sustainable growth in digital commerce.

This project proposes the development of a **Secure Escrow and E-Commerce Mediation System**, a web-based platform that acts as a trusted third party between buyers and sellers. The system temporarily holds buyer payments in escrow and releases funds only after delivery confirmation. By integrating transaction management, dispute resolution, and role-based access control, the platform ensures transparency, security, and accountability in online transactions.

**Problem Statement**

Online transactions in Ethiopia lack a structured escrow mechanism. Payments are transferred directly to sellers without protection, and disputes are handled informally. This results in high fraud risk, delivery failures, and payment conflicts. A secure mediation system is needed to protect users and strengthen trust in digital marketplaces.

**Objectives**

**General Objective:**
To design and develop a secure escrow-based system that safeguards online transactions and promotes trust.

**Specific Objectives:**

1. Develop secure user management for buyers, sellers, and administrators.
2. Implement an escrow wallet to hold payments during transactions.
3. Design transaction tracking with defined states (Pending, Paid, Shipped, Completed, Disputed).

**Example Scenario**

A buyer deposits payment into escrow. The seller ships the product. After delivery confirmation, the system releases payment to the seller. If a dispute occurs, the administrator reviews the case and determines whether to refund or release funds.

**Conclusion**

The proposed system enhances trust, reduces fraud, and formalizes online transactions by introducing a secure escrow and mediation mechanism.

## **Project Title 2**: **National Internship & Job Matching Platform**

**Introduction**

In Ethiopia, students and universities face challenges connecting with employers for internships and jobs. Students often miss opportunities, while employers struggle to find suitable candidates. This project proposes a **web-based platform** that connects students, universities, and companies, enabling skill profiling, internship tracking, and intelligent match-making for efficient career placement.

**Problem Statement**

Currently, universities in Ethiopia lack a standardized system to monitor student internships and employment readiness. Students struggle to discover opportunities aligned with their skills, and employers cannot easily identify suitable candidates. This leads to mismatched placements, underutilized talent, and lost opportunities for both students and companies. A national platform is needed to bridge this gap and streamline internship and job matching.

**Objectives**

**General Objective:**

To design and develop a national platform that efficiently connects students, universities, and employers for internships and job opportunities, promoting career readiness and workforce alignment.

**Specific Objectives:**

1. Develop secure user management for students, universities, and employers.
2. Create student skill profiles and enable integration with university systems.
3. Implement AI-driven match-making between candidates and available internships or jobs.
4. Track internships and collect feedback for both students and universities.
5. Provide employers with dashboards for managing postings and generating reports.

**Example Scenario**

A student creates a skill profile, which is verified by their university. The system recommends relevant internships and job openings using AI-based matching. The student applies to an opportunity, and the employer reviews applications on their dashboard. Universities can track student placements and collect feedback on internship performance. If issues arise during an internship, the platform provides a structured reporting mechanism to resolve disputes or provide guidance.

**Conclusion**

The platform improves student employability, strengthens university-employer collaboration, and ensures effective internship and job matching nationwide.

# Project Title 3: ConfiguraFlow – Dynamic, Configurable Business Management Platform

## Introduction

Small and medium businesses in Ethiopia often struggle with software that doesn't fit their needs. Industry-specific solutions are expensive, rigid, and hard to maintain. ConfiguraFlow offers a **single, unified platform** that can be configured for any business—pharmacy, restaurant, salon, retail, or gym—without writing code. The platform enables implementers and end-users to shape workflows, data, UI, and rules through configuration, making business management accessible, flexible, and affordable.

## Problem Statement

Traditional approaches—custom development, off-the-shelf, or separate industry solutions—are either too costly, inflexible, or redundant. Businesses need software that fits their processes, adapts as they grow, and is affordable to deploy and maintain.

## Objectives

### General Objective:

To develop a configurable business management platform that serves multiple industries from a single codebase.

### Specific Objectives:

1. Enable dynamic entity and workflow management for any business process.
2. Provide a visual UI builder for dashboards, forms, and reports.
3. Implement a rule engine for validation, automation, and calculations.
4. Support multi-tenant architecture with secure, isolated business data.
5. Offer user and role management for implementers and end-users.

## Example Scenario

A consultant configures a pharmacy system by defining medicine entities, workflows for order processing, and dashboard widgets. The pharmacy staff uses the system daily to manage inventory, process orders, and generate reports. If the business evolves, the implementer or admin can update workflows, forms, or rules without coding.

## Conclusion

ConfiguraFlow transforms how business software is delivered. One platform serves infinite business models through configuration, reducing cost, speeding deployment, and providing tailored solutions. Businesses get adaptable software, implementers save development effort, and end-users enjoy intuitive interfaces built for their workflows.